



DATA ANALYSIS PORTFOLIO

By Armani Chalmers



Professional Background

I'm a detail-oriented aspiring Data Analyst looking to further my knowledge and experience.

I'm in pursuit of a new career in data analytics. Holding a Bachelors Degree in Music Business, I have recently pivoted into tech for better opportunities.

In my last position, I worked with Salesforce and I'm working on diving deeper into the analytical side in the future.

I've recently stepped into a new position working with data such as collecting, importing, and cleaning. I bring a set of valuable talents such as my analytical mind, organization, and strategic thinking. I have a knack for problem-solving and research.

I've been working on upskilling not only with data analytics, but also with project management skills to help my organization and leadership skills. I recently obtained a ScrumMaster Certification, a Jira Fundamentals badge, as well as gained more knowledge in Agile framework. I'm highly dedicated to growing as a data professional and will continue to learn more in hopes of landing the right job opportunity.

Thank you for your time and consideration.

Thank you 

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Udemy Project Description/Business Problem

As a Data Analyst for Udemy, I was given 3 weeks to present data on how and where we can increase revenue in order to increase quarterly earnings. I collected data such as course subjects, price per course (whether they were free or paid), number of subscribers, and course level,

This report will show which courses and subjects are the least and most popular. With Web Development being the most popular subject, The findings will determine which of its levels' prices can be increased most. The data will also determine which subjects and levels students are most satisfied with based on ratings.

This data is represented with pie charts and bar charts.

Data Design

Data Cleaning:

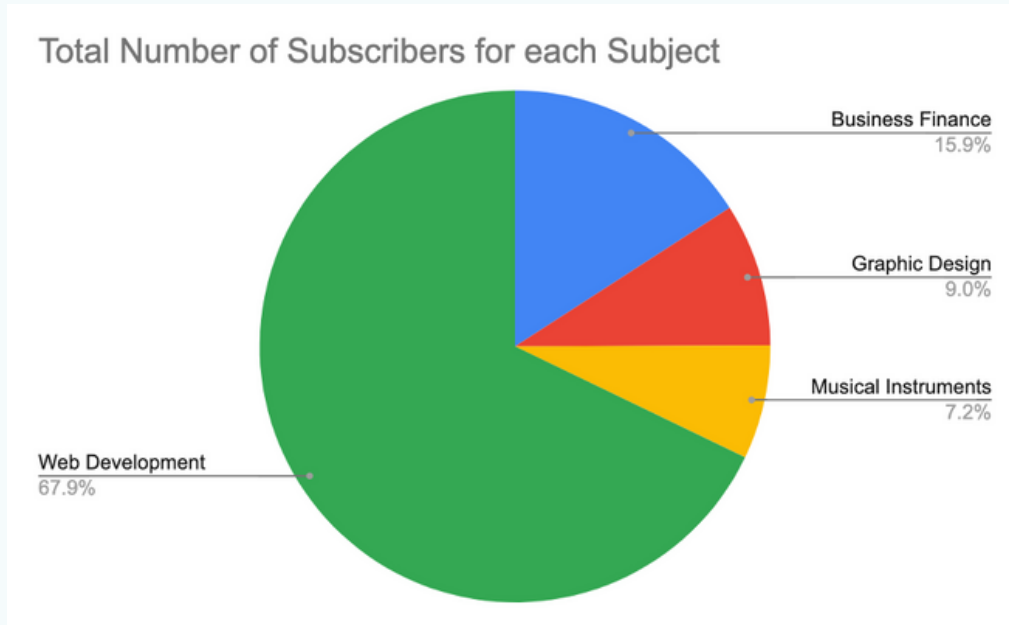
To clean the data, duplicates and blanks were removed. All headers, as well as the Web Development subject title were reformatted to keep consistent with the other headers and subject formats.

Data Visualization:

Pivot tables and charts were created in Google Sheets and Tableau to visualize and support the data findings and answer the business problem.

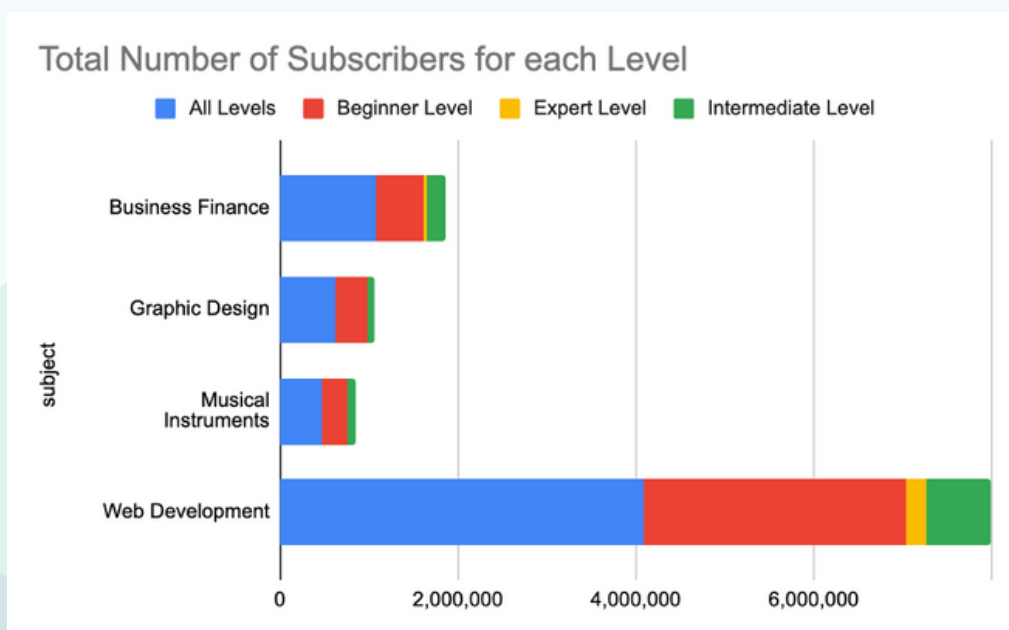
Findings

Figure 1: Total Number of Subscribers for each Subject (Google Sheets)



Web Development has the highest number of subscribers, showing us that it is the most popular subject.

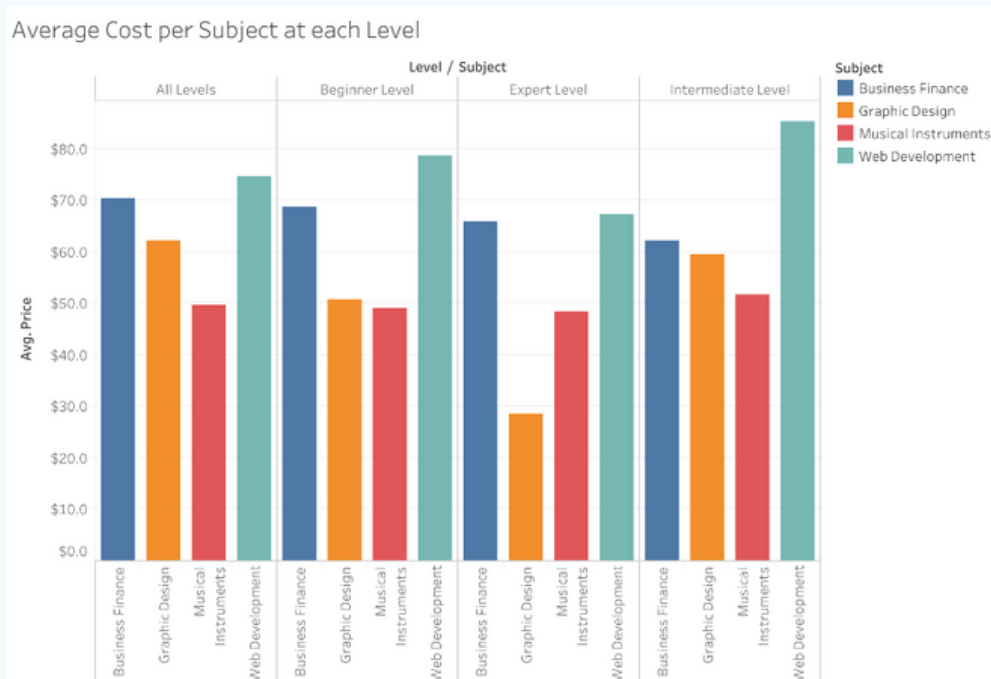
Figure 2: Total Number of Subscribers for each Level (Google Sheets)



The stacked bar chart above determines that the courses teaching all levels of Web Development have the highest number of subscribers.

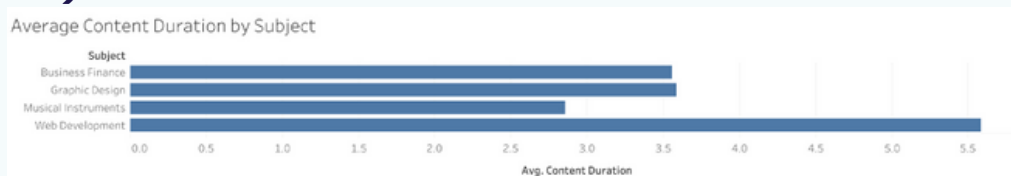
Findings Cont'd.

Figure 3: Average Cost per Subject at each Level (Tableau)



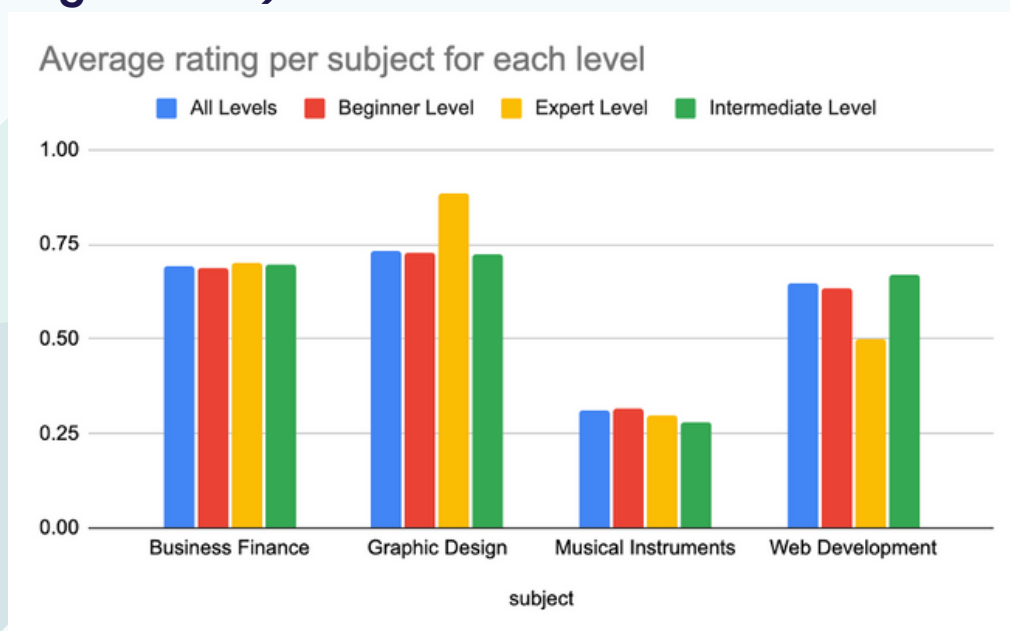
The bar column bar chart above determines that Web Development at the intermediate level has the highest averaged cost and Graphic Design at the expert level has the lowest.

Figure 4: Average Content Duration by Subject (Tableau)



Web Development has the highest averaged content duration.

Figure 5: Average Rating per Subject for each Level (Google Sheets)



Graphic Design has the highest ratings, especially the expert level courses.

Analysis



Root Cause Analysis is the process used to find what the root cause/underlying problem is in order to figure out how to solve the problem.

- Why is Web Development driving in the most revenue?
- Why is Web Development the most popular subject?
- Why doesn't Web Development have the highest ratings?
- Why is Musical Instruments driving in the least revenue?
- Why is Musical Instruments the least popular subject?

Based on the data and visualization findings, Web Development is the most popular subject for courses. Business Finance is the second most popular, followed by Graphic Design and Musical Instruments. Although Web Development is the most popular, the findings show that it does not have the highest ratings on average and that may be due to the price points and/or quality of content. However, Web Development has the highest content duration, and this is what is driving user purchases and retention. Furthermore, the findings show that the expert level across all subjects has the lowest average cost. All levels of each subject have the highest number of subscribers, indicating that our users find more value in learning the subjects at all levels instead of specific levels.

Conclusion

The data concludes that a slight price increase in Web Development will generate revenue growth for the company due to its largest number of subscribers overall. The price point can also be slightly increased across all subjects at the expert level as well as better quality content to drive up the ratings. Web Development will continue to lead due to the current popularity of tech so we must commit to making sure that we are delivering quality content so that our users can find value in it. It is highly recommended to increase Web Development prices in order to increase the business' revenue.



Travel+Leisure World's Best Hotels 2022 Project Background

This project aims to provide data that will show us what type of hotel theme and region was most popular in 2022 in order to give us an idea of what types of hotels should be developed in the upcoming years.

Region, score, rank, and theme were main data collected to support the business problem. The data shows us the score/rank of top 100 best hotels by regions and themes from most to least popular in 2022.

This data is represented with pivot tables and bar charts.

Data Design

Data Cleaning:

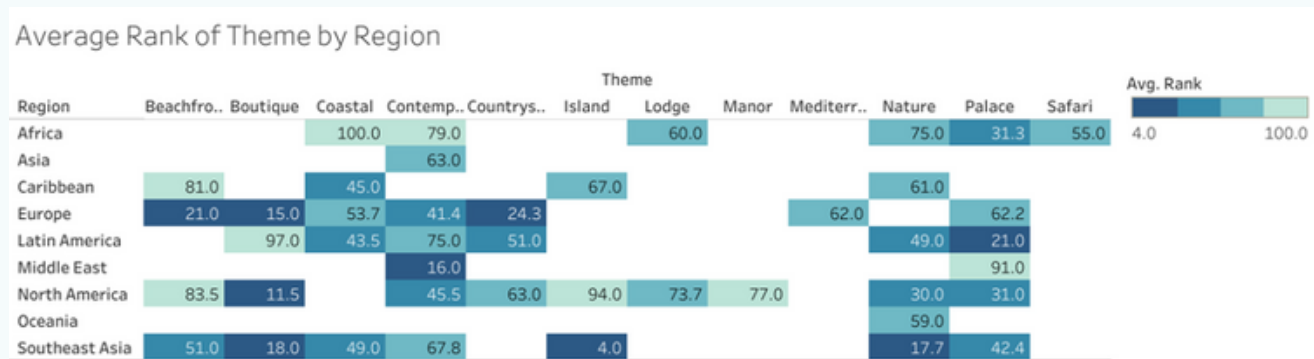
To clean the data, I checked for duplicates (none were found) and blanks were removed. Headers were reformatted to keep consistent. Found and replaced data under "Listed_2021" to "Yes" and "No".

Data Visualization:

Pivot tables and charts were created in Google Sheets and Tableau to visualize and support the data findings and answer the problem.

Findings

Table 1: Average Rank of Theme by Region (Tableau)

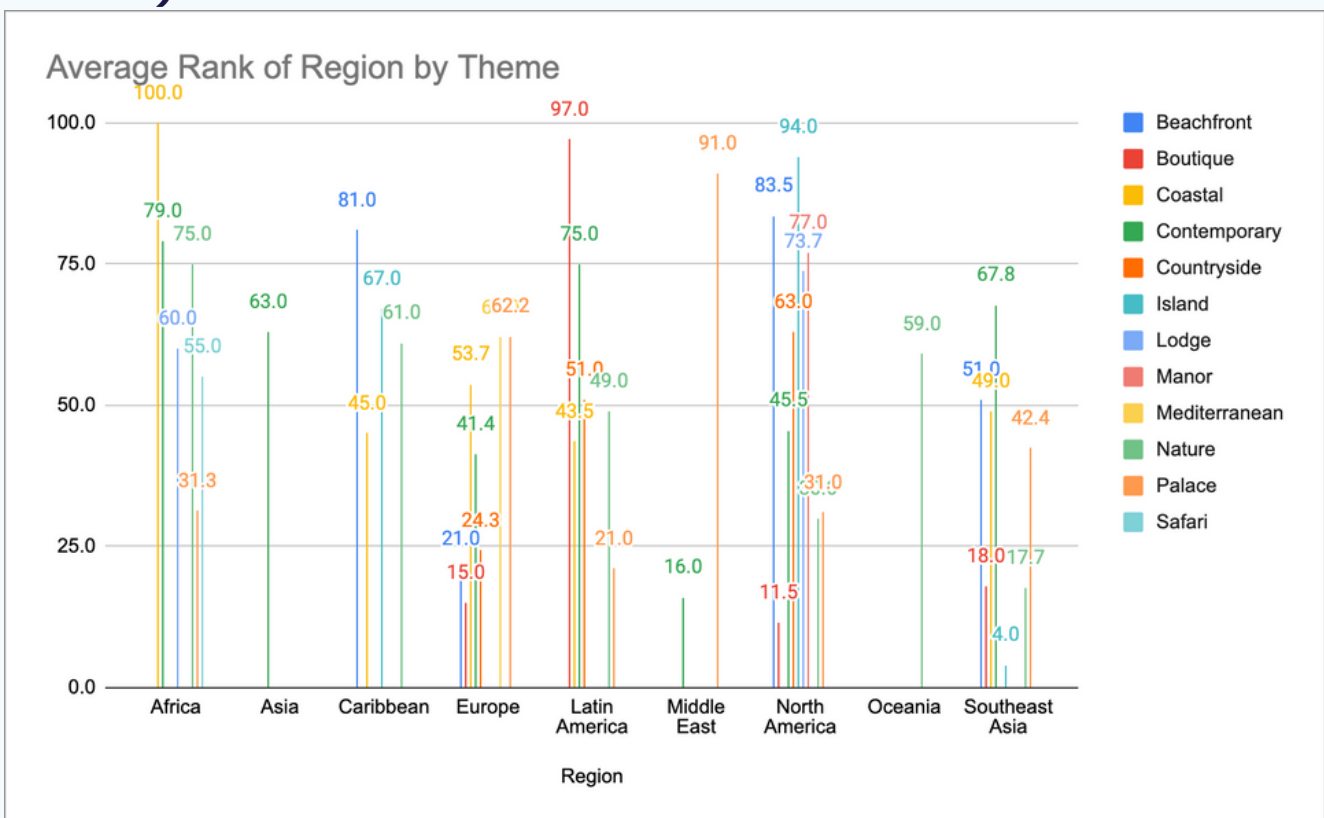


(Most popular = Dark blue, Least popular = Light blue)

The most popular theme by region is Southeast Asia Island averaging at #4 rank.

African Coastal comes in at the bottom at #100.

Figure 1: Average Rank of Region by Theme (Google Sheets)

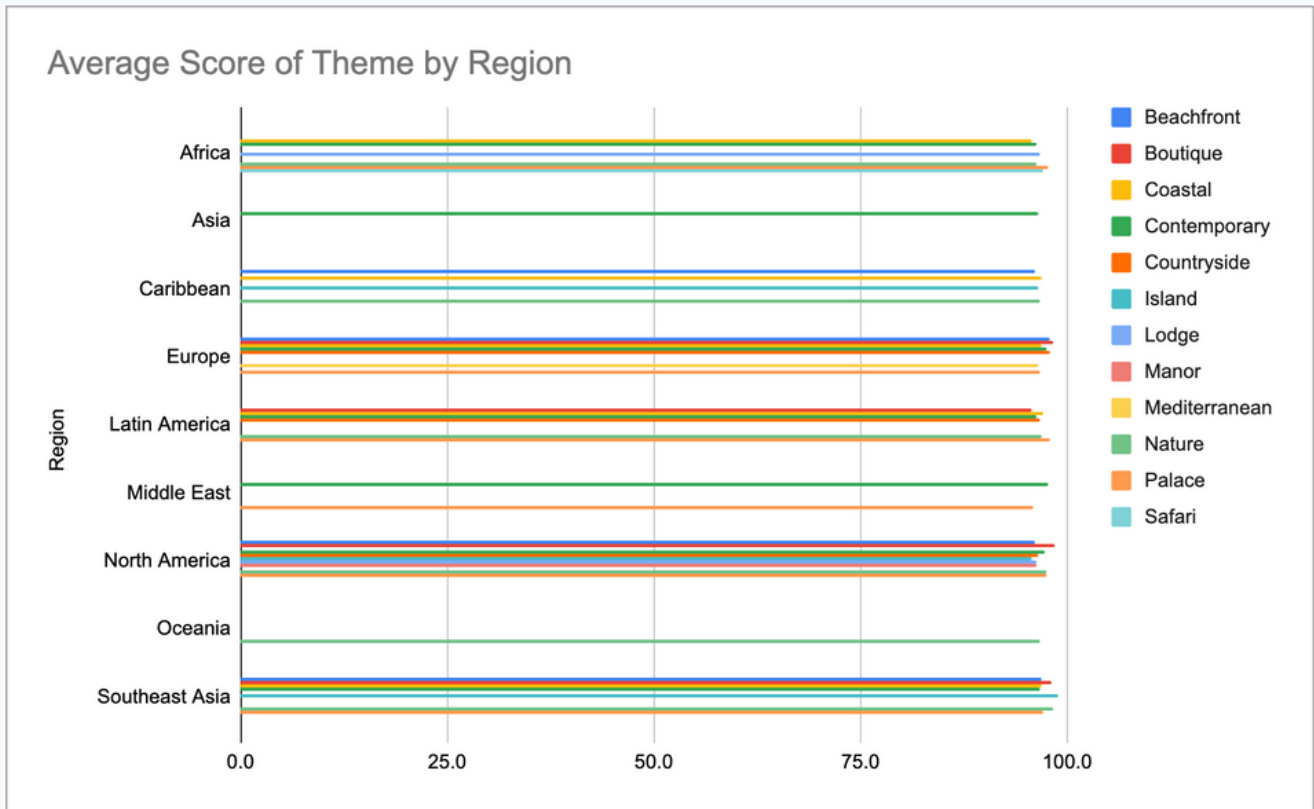


(Lowest number = Most Popular, Highest number = Least Popular)

A bar chart representation of the table above showing Southeast Asia Island at 4.0 and Africa Coastal at 100.0.

Findings Cont'd.

Figure 2: Average Score of Theme by Region (Google Sheets)



Average theme by region shown by score. They are all very close but Southeast Asia Island takes the lead at an average score of 99.

Findings Cont'd.

Table 2: Average Score of Theme by Region (Google Sheets)

Theme	Region	AVERAGE of Score
Beachfront	Caribbean	96.2
	Europe	97.9
	North America	96.2
	Southeast Asia	96.9
Beachfront Total		96.7
Boutique	Europe	98.3
	Latin America	95.8
	North America	98.5
	Southeast Asia	98.2
Boutique Total		97.9
Coastal	Africa	95.7
	Caribbean	97.0
	Europe	97.0
	Latin America	97.1
	Southeast Asia	96.9
Coastal Total		96.9
Contemporary	Africa	96.3
	Asia	96.6
	Europe	97.5
	Latin America	96.4
	Middle East	97.7
	North America	97.3
	Southeast Asia	96.7
Contemporary Total		97.1
Countryside	Europe	97.9
	Latin America	96.8
	North America	96.6
Countryside Total		97.5
Island	Caribbean	96.5
	North America	95.8
	Southeast Asia	99.0
Island Total		97.6
Lodge	Africa	96.6
	North America	96.3
Lodge Total		96.4
Manor	North America	96.3
Manor Total		96.3
Mediterranean	Europe	96.6
Mediterranean Total		96.6
Nature	Africa	96.4
	Caribbean	96.6
	Latin America	97.0
	North America	97.5
	Oceania	96.6
	Southeast Asia	98.2
Nature Total		97.2
Palace	Africa	97.7
	Europe	96.6
	Latin America	97.9
	Middle East	95.9
	North America	97.5
	Southeast Asia	97.2
Palace Total		97.1
Safari	Africa	97.0
Safari Total		97.0

This pivot table shows each region and theme along with their averaged scores, as well as total average score for each theme.

Analysis

Root Cause Analysis is the process used to find what the root cause/underlying problem is in order to figure out how to solve the problem.

- Why are Southeast Asian Island themed hotels most popular?
- Why were they ranked this way?
- Why were they scored this way?

Based on the findings, Southeast Asian Island themed hotels were ranked and scored most popular, followed by North American Boutique and European Boutique. Top scores were determined by analyzing the highest averaged numbers whereas, top rank was determined by analyzing the lowest averaged numbers. Both score and rank were determined by the ratio of Theme and Region. The findings also determined that African Coastal, Latin American Boutique, and North American Island themed hotels were least popular amongst the Top 100 World's Best Hotels list.

Conclusion

The findings determine that the next hotels developed should be Southeast Asian Island, North American Boutique, and European Boutique themed hotels. These specific regions and themes were ranked and scored the highest according to the Travel+Leisure World's Best Hotels 2022 list. This report also provides data that supports that African Coastal, Latin American Boutique, and North American Island themed hotels should be steered away from. It is highly recommended to develop Southeast Asian Island themed hotels, first and foremost, and then North American Boutique and European Boutique themed hotels.

Appendix



Google Sheets Data Set for Udemy Project

https://docs.google.com/spreadsheets/d/1jubi72QJhooJCySTuP11-GFzgG2YOs_BITlbWzd9dMs/edit#gid=1611635922

Tableau Data Visualization For Udemy Project

https://public.tableau.com/views/EntryLevelUdemyProject/Dashboard1?:language=en-US&:display_count=n&:origin=viz_share_link

Google Sheets Data Set for Travel+Leisure World's Best Hotels 2022 Project

<https://docs.google.com/spreadsheets/d/1lhDWAKgxIBnTgIM2sLmGn-z26VbV2Wep4xim2XADLUc/edit#gid=0>

Tableau Data Visualization For Travel+Leisure World's Best Hotels 2022 Project

https://public.tableau.com/views/entryleveltop100hotels/Dashboard1?:language=en-US&:display_count=n&:origin=viz_share_link